

Reciprocity and minimal degree of $\text{tr}_{V^\lambda}(\Omega)$ for the staircase product $\Omega = \prod_{\text{red}(w_0)} (T_i + 1)$ in $H_n(q)$

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Abstract

Let $H_n(q)$ be the Iwahori–Hecke algebra of S_n with quadratic relation $(T_i - q)(T_i + 1) = 0$, and let $\Omega = \prod_{k=2}^n \prod_{j=k-1}^1 (T_j + 1)$ be the product of $T_i + 1$ over the staircase reduced word for the longest element w_0 (length $\binom{n}{2}$). For $\lambda \vdash n$ write $\chi^\lambda(\Omega) = \text{tr}_{V^\lambda}(\Omega) \in \mathbb{Z}[q]$. We prove **(A) Reciprocity**: $\chi^\lambda(\Omega)$ is self-reciprocal of span $\binom{n}{2}$, $\chi^\lambda(\Omega)(q^{-1}) = q^{-\binom{n}{2}} \chi^\lambda(\Omega)(q)$. The proof is short and complete: the Kazhdan–Lusztig bar involution sends $\Omega \mapsto q^{-\binom{n}{2}} \Omega$, and every Specht/Hoefsmit module V^λ is bar-self-dual (proved here by specialisation at $q = 1$). For **(B) Minimal degree** we prove the exact reduction $\min \deg_q \chi^\lambda(\Omega) = (\text{minimum cost of a closed walk on } \text{SYT}(\lambda) \text{ along the staircase word})$, where the cost rule is read off the $q \rightarrow 0$ orders of the Hoefsmit matrices and “no cancellation” is guaranteed by Hoefsmit positivity. We prove the upper bound $\min \deg_q \chi^\lambda(\Omega) \leq n(\lambda)$ for every λ with at most one part equal to 1 (in particular all two-row shapes), via the *all-stay* walk at the row-superstandard tableau, and we identify the matching lower bound as a sharp combinatorial lemma (the “Core Lemma”), verified exactly on all 29 non-vanishing shapes with $n \leq 7$. The whole package pins down the exact shape of the trace polynomial whose vanishing the τ -dichotomy detects: palindromic, span $\binom{n}{2}$, bottom term $q^{n(\lambda)}$.

1 Setup and statement

1.1 The algebra and the element Ω

Let $H_n(q)$ be the Iwahori–Hecke algebra of the symmetric group S_n over $K = \mathbb{Q}(q)$, with generators $T_1, \dots, T_{n-1}$, braid relations $T_i T_{i+1} T_i = T_{i+1} T_i T_{i+1}$, $T_i T_j = T_j T_i$ ($|i - j| \geq 2$), and the quadratic relation

$$(T_i - q)(T_i + 1) = 0, \quad \text{equivalently } T_i^2 = (q - 1)T_i + q. \quad (1)$$

Over K the algebra is split semisimple, with irreducible modules V^λ ($\lambda \vdash n$), $\dim V^\lambda = f^\lambda = \#\text{SYT}(\lambda)$; write $\rho^\lambda: H_n(q) \rightarrow \text{Mat}_{f^\lambda}(K)$ and $\chi^\lambda(x) = \text{tr } \rho^\lambda(x)$.

The *staircase word* is the reduced word $\mathbf{w}_0 = (1)(2\ 1)(3\ 2\ 1) \cdots (n-1 \cdots 1)$ for the longest element $w_0 \in S_n$, of length $L = \binom{n}{2}$. We set

$$\Omega = \prod_{\text{letters } i \text{ of } \mathbf{w}_0, \text{ in order}} (T_i + 1) = (T_1 + 1)(T_2 + 1)(T_1 + 1)(T_3 + 1)(T_2 + 1)(T_1 + 1) \cdots \in H_n(q). \quad (2)$$

For a partition $\lambda \vdash n$ put $n(\lambda) = \sum_{i \geq 1} (i - 1)\lambda_i = \sum_{j \geq 1} \binom{\lambda'_j}{2}$, where λ' is the conjugate.

1.2 Main results

Theorem 1 (Reciprocity). *For every $\lambda \vdash n$,*

$$\chi^\lambda(\Omega)(q^{-1}) = q^{-\binom{n}{2}} \chi^\lambda(\Omega)(q).$$

Equivalently $\chi^\lambda(\Omega) \in \mathbb{Z}[q]$ is self-reciprocal (palindromic) with $\min \deg_q + \max \deg_q = \binom{n}{2}$.

Theorem 2 (Minimal degree, reduction). *Assume $\chi^\lambda(\Omega) \neq 0$. Then*

$$\min \deg_q \chi^\lambda(\Omega) = \text{mincost}(\lambda),$$

the minimum, over all closed walks on $\text{SYT}(\lambda)$ along the staircase word, of the walk cost defined in §3. By Theorem 1, $\max \deg_q \chi^\lambda(\Omega) = \binom{n}{2} - \text{mincost}(\lambda)$.

Theorem 3 (Minimal degree, upper bound). *If λ has at most one part equal to 1 (in particular, for every two-row shape $\lambda = (a, b)$), then $\text{mincost}(\lambda) \leq n(\lambda)$, and hence $\min \deg_q \chi^\lambda(\Omega) \leq n(\lambda)$.*

Conjecture 4 (Core Lemma; lower bound). *For every λ with $\chi^\lambda(\Omega) \neq 0$, $\text{mincost}(\lambda) \geq n(\lambda)$. Consequently $\min \deg_q \chi^\lambda(\Omega) = n(\lambda)$ and $\max \deg_q \chi^\lambda(\Omega) = \binom{n}{2} - n(\lambda)$.*

Conjecture 4 is verified *exactly* ($\text{mincost} = n(\lambda) = \text{true min deg}$) on all 29 non-vanishing shapes with $n \leq 7$ (§6). Together with Theorem 3 it gives the clean structural statement: when nonzero, $\chi^\lambda(\Omega) = q^{n(\lambda)}(1+q)^b P_\lambda(q)$ with P_λ a positive palindrome.

2 Proof of Theorem 1 (Reciprocity)

2.1 The bar involution sends Ω to $q^{-\binom{n}{2}}\Omega$

Let $\sigma: K \rightarrow K$ be the field automorphism fixing $\mathbb{Q}$ with $\sigma(q) = q^{-1}$ (well defined, as q^{-1} is again transcendental and generates K); note $\sigma^2 = \text{id}$.

Lemma 5. *There is a unique ring automorphism $b: H_n(q) \rightarrow H_n(q)$ that is σ -semilinear over $\mathbb{Z}[q, q^{-1}]$ ($b(fx) = \sigma(f)b(x)$), satisfies $b(T_i) = T_i^{-1}$, and is involutive ($b^2 = \text{id}$). Moreover*

$$b(T_i + 1) = q^{-1}(T_i + 1), \quad \text{hence} \quad b(\Omega) = q^{-\binom{n}{2}}\Omega.$$

Proof. Existence. By the universal property of $H_n(q)$ it suffices that the images T_i^{-1} satisfy the defining relations after applying σ to the scalars. From (1) the eigenvalues of T_i are $q, -1$, so those of T_i^{-1} are $q^{-1}, -1$, i.e. T_i^{-1} satisfies $(T_i^{-1} - q^{-1})(T_i^{-1} + 1) = 0$, which is exactly σ applied to (1). The braid relations are preserved by $T_i \mapsto T_i^{-1}$ (invert both sides of $T_i T_{i+1} T_i = T_{i+1} T_i T_{i+1}$ and use that the relation is symmetric). Thus a σ -semilinear ring homomorphism b with $b(T_i) = T_i^{-1}$ exists. It is involutive on generators and on scalars, hence $b^2 = \text{id}$, so b is an automorphism. Uniqueness is clear since the T_i generate.

Computation. Inverting (1): from $T_i^2 = (q-1)T_i + q$ we get $T_i^{-1} = q^{-1}T_i - 1 + q^{-1}$ (indeed $T_i(q^{-1}T_i - 1 + q^{-1}) = q^{-1}((q-1)T_i + q) - T_i + q^{-1}T_i = 1$). Hence

$$b(T_i + 1) = T_i^{-1} + b(1) = q^{-1}T_i - 1 + q^{-1} + 1 = q^{-1}(T_i + 1).$$

Since b is a ring homomorphism and Ω is a product of $\binom{n}{2}$ factors $T_{i_k} + 1$, $b(\Omega) = \prod_k b(T_{i_k} + 1) = \prod_k q^{-1}(T_{i_k} + 1) = q^{-\binom{n}{2}}\Omega$. $\square$

2.2 Bar-self-duality of V^λ

Lemma 6. *For every $\lambda \vdash n$ and every $h \in H_n(q)$,*

$$\chi^\lambda(b(h))(q^{-1}) = \chi^\lambda(h)(q).$$

Proof. Fix a K -basis of V^λ and let σ act entrywise on $\text{Mat}_{f^\lambda}(K)$ (a ring homomorphism). Define $\tilde{\rho} := \sigma \circ \rho^\lambda \circ b$. This is a genuine K -linear representation of $H_n(q)$: it is multiplicative because b , ρ^λ and entrywise σ all are, and it is K -linear because for $f \in K$, $\tilde{\rho}(fh) = \sigma(\rho^\lambda(b(fh))) = \sigma(\rho^\lambda(\sigma(f)b(h))) = \sigma(\sigma(f)\rho^\lambda(b(h))) = \sigma^2(f)\tilde{\rho}(h) = f\tilde{\rho}(h)$, using $\sigma^2 = \text{id}$. Its character is

$$\tilde{\chi}(h)(q) = \sigma(\chi^\lambda(b(h)))(q) = \chi^\lambda(b(h))(q^{-1}). \quad (3)$$

Since $H_n(q)$ is split semisimple, $\tilde{\rho} \cong \bigoplus_\mu (V^\mu)^{\oplus m_\mu}$ with $m_\mu \in \mathbb{Z}_{\geq 0}$ and $\tilde{\chi} = \sum_\mu m_\mu \chi^\mu$. We determine the m_μ by specialising at $q = 1$. Recall (Geck–Pfeiffer) that the irreducible character values are Laurent polynomials, $\chi^\mu(T_w) \in \mathbb{Z}[q, q^{-1}]$, and that specialisation $q \mapsto 1$ identifies $H_n(q) \rightsquigarrow \mathbb{Q}[S_n]$ with $\chi^\mu(T_w)|_{q=1} = \chi^{S^\mu}(w)$, the Specht character. Evaluate (3) on T_w at $q = 1$. At $q = 1$ we have $T_i^2 = 1$, so $b(T_i) = T_i^{-1} = T_i$ specialises to the identity automorphism of $\mathbb{Q}[S_n]$, and $\sigma|_{q=1} = \text{id}$. Writing $b(T_w) = \sum_x c_{w,x}(q) T_x$ with $c_{w,x} \in \mathbb{Z}[q, q^{-1}]$ and $\sum_x c_{w,x}(1) x = w$ (image under the identity), we get

$$\tilde{\chi}(T_w)(1) = \chi^\lambda(b(T_w))(1) = \sum_x c_{w,x}(1) \chi^{S^\lambda}(x) = \chi^{S^\lambda}(w).$$

Thus $\tilde{\chi}|_{q=1} = \chi^{S^\lambda} = \sum_\mu m_\mu \chi^{S^\mu}$. By linear independence of the irreducible S_n -characters, $m_\lambda = 1$ and $m_\mu = 0$ otherwise; hence $\tilde{\chi} = \chi^\lambda$ identically. Combined with (3) this is the claim. $\square$

Remark 7. Lemma 6 is the statement that V^λ is isomorphic to its bar-twist $\sigma^* b^* V^\lambda$. For type A this is the well known bar-self-duality of cell modules in Kazhdan–Lusztig theory; the specialisation proof above is self-contained.

2.3 Conclusion of Theorem 1

Write $P(q) = \chi^\lambda(\Omega)(q)$. Apply Lemma 6 with $h = \Omega$ and then Lemma 5:

$$P(q) = \chi^\lambda(\Omega)(q) = \chi^\lambda(b(\Omega))(q^{-1}) = \chi^\lambda(q^{-\binom{n}{2}}\Omega)(q^{-1}).$$

As χ^λ is K -linear, $\chi^\lambda(q^{-\binom{n}{2}}\Omega)(q) = q^{-\binom{n}{2}}P(q)$ as a function of q ; substituting $q \mapsto q^{-1}$ gives $\chi^\lambda(q^{-\binom{n}{2}}\Omega)(q^{-1}) = q^{\binom{n}{2}}P(q^{-1})$. Therefore $P(q) = q^{\binom{n}{2}}P(q^{-1})$, i.e. $P(q^{-1}) = q^{-\binom{n}{2}}P(q)$. $\square$

3 Proof of Theorem 2 (reduction to a walk)

3.1 Hoefsmit seminormal form and the $q \rightarrow 0$ orders

We use the Mathas/Murphy seminormal form: V^λ has basis $\{v_T\}_{T \in \text{SYT}(\lambda)}$, and for the generator T_i , with $d = d_T(i) := c(i+1) - c(i)$ the axial distance in T ($c(\text{cell}(r, c)) = c - r$, rows/cols 0-indexed):

- $d = 1$ ($i, i+1$ same row): $T_i v_T = q v_T$;
- $d = -1$ ($i, i+1$ same column): $T_i v_T = -v_T$;

- $|d| \geq 2$: on $\text{span}(v_{T^+}, v_{T^-})$, where T^+ is the member of $\{T, s_i T\}$ with $i+1$ at the higher content ($d_{T^+} = +|d|$) and $T^- = s_i T^+$, the matrix of T_i is

$$\begin{pmatrix} A_{|d|} & A_{|d|}A_{-|d|} + q \\ 1 & A_{-|d|} \end{pmatrix}, \quad A_d = \frac{q-1}{1-q^{-d}} = \frac{(q-1)q^d}{q^d-1}.$$

(Trace $A_{|d|} + A_{-|d|} = q-1$, determinant $-q$, so eigenvalues $q, -1$.)

Hence the matrix $M := T_i + 1$ on the block $\{T^+, T^-\}$ (rows/columns ordered T^+, T^-) is

$$M = \begin{pmatrix} A_{|d|} + 1 & A_{|d|}A_{-|d|} + q \\ 1 & A_{-|d|} + 1 \end{pmatrix},$$

while $Mv_T = (q+1)v_T$ in the same-row case and $Mv_T = 0$ in the same-column case.

Lemma 8 ($q \rightarrow 0$ orders). *Write ord for the order of vanishing at $q = 0$. In the block above ($|d| \geq 2$):*

$$\text{ord}(A_{|d|} + 1) = 0, \quad \text{ord}(1) = 0, \quad \text{ord}(A_{|d|}A_{-|d|} + q) = 1, \quad \text{ord}(A_{-|d|} + 1) = 1.$$

Consequently the column of M indexed by T^+ has both entries of order 0, and the column indexed by T^- has both entries of order ≥ 1 . In the same-row case the (diagonal) entry $q+1$ has order 0; the same-column case gives the zero entry.

Proof. Near $q = 0$: $A_{|d|} = \frac{(q-1)q^{|d|}}{q^{|d|}-1} = q^{|d|} + O(q^{|d|+1})$, so $A_{|d|} + 1 = 1 + O(q^{|d|})$ (order 0). Next $A_{-|d|} = \frac{q-1}{1-q^{|d|}} = -1 + q + O(q^{|d|})$, so $A_{-|d|} + 1 = q + O(q^2)$ (order 1). Finally $A_{|d|}A_{-|d|} = \frac{-(q-1)^2 q^{|d|}}{(q^{|d|}-1)^2} = -q^{|d|} + O(q^{|d|+1})$, so $A_{|d|}A_{-|d|} + q = q + O(q^2)$ (order 1, as $|d| \geq 2$). The entry 1 and $q+1$ have order 0. $\square$

3.2 Walks and cost

Expanding the matrix product $\rho^\lambda(\Omega) = \prod_{k=1}^L \rho^\lambda(T_{i_k} + 1)$, the diagonal entry $p_T(q) := \rho^\lambda(\Omega)_{T,T}$ is the sum over sequences $T = U_0, U_1, \dots, U_L = T$ of $\prod_{k=1}^L M_{U_{k-1}, U_k}^{(k)}$, where $M^{(k)} = \rho^\lambda(T_{i_k} + 1)$. Each step either keeps the tableau ($U_k = U_{k-1}$) or swaps to $U_k = s_{i_k} U_{k-1}$ (only when $|d| \geq 2$, so the result is standard). We call such a sequence a *closed walk along the staircase word starting at T* .

Definition 9 (cost). The *cost* of step k in a walk is, by definition, the order at $q = 0$ of the entry $M_{U_{k-1}, U_k}^{(k)}$. By Lemma 8 the order of an entry depends only on its *column* index U_k :

$$\text{cost}_k = \begin{cases} 0 & \text{if } d_{U_k}(i_k) = +1 \text{ } (i_k, i_k+1 \text{ same row in } U_k), \\ 0 & \text{if } d_{U_k}(i_k) \geq 2 \text{ } (U_k = T^+ : i_k+1 \text{ at higher content}), \\ 1 & \text{if } d_{U_k}(i_k) \leq -2 \text{ } (U_k = T^- : i_k+1 \text{ at lower content}). \end{cases}$$

The case $d_{U_k}(i_k) = -1$ (same column in U_k) cannot occur: the column of $M^{(k)}$ indexed by such a tableau is identically zero (the same-column generator entry), so no walk visits it. Each step is a *stay* ($U_{k-1} = U_k$) or a *swap* ($U_{k-1} = s_{i_k} U_k$, possible only when $|d_{U_k}(i_k)| \geq 2$). In words: step k costs 1 iff in the tableau U_k the value i_k+1 has strictly lower content than i_k . The cost of a walk is $\sum_k \text{cost}_k$, and $\text{mincost}(\lambda)$ is the minimum cost over all closed walks (over all starting T).

Remark 10. The cost is read off the *column* of each factor because that is what the $q \rightarrow 0$ order of $M_{U_{k-1}, U_k}^{(k)}$ depends on (Lemma 8). For a *stay* step $U_{k-1} = U_k$ the column and the current state coincide, so the rule reads “ i_k+1 lower content than i_k in the current tableau”; this is the only case used in the upper bound §4. The symmetric $q \rightarrow \infty$ computation reads the cost off the *row* index instead and returns $\binom{n}{2} - \max \deg$; by Theorem 1 the two minima agree, and both equal the true min deg on every shape tested (§6).

3.3 No cancellation, via Hoefsmit positivity

Lemma 11. $\min \deg_q \chi^\lambda(\Omega) = \min_T \text{ord}_{q=0} p_T(q) = \text{mincost}(\lambda)$.

Proof. By the Hoefsmit positivity lemma (Lemma 1 of the $b' = 1$ analysis; arXiv companion), every entry of $\rho^\lambda(T_i + 1)$ lies in $\mathbb{Q}_{\geq 0}(q)$: it is a ratio of polynomials with non-negative coefficients. (For $|d| \geq 2$ two of the block entries simplify cleanly to ratios of q -integers,

$$A_{|d|} + 1 = \frac{q^{|d|+1} - 1}{q^{|d|} - 1} = \frac{[|d| + 1]_q}{[|d|]_q}, \quad A_{-|d|} + 1 = \frac{q(1 - q^{|d|-1})}{1 - q^{|d|}} = q \frac{[|d| - 1]_q}{[|d|]_q},$$

both in $\mathbb{Q}_{\geq 0}(q)$; the entry 1 is positive; the remaining off-diagonal entry $A_{|d|}A_{-|d|} + q$ is positive by [[hoefsmit-positivity]], see Remark 12.) Consequently each diagonal entry $p_T(q)$, being a sum of products of such non-negative ratios, again lies in $\mathbb{Q}_{\geq 0}(q)$, and so does $\chi^\lambda(\Omega) = \sum_T p_T$. A finite sum of elements of $\mathbb{Q}_{\geq 0}(q)$ cannot have a cancellation in its lowest-order term: writing each $p_T = q^{m_T} u_T(q)$ with $u_T \in \mathbb{Q}_{\geq 0}(q)$, $u_T(0) > 0$, the lowest order of the sum is $\min_T m_T$ and its coefficient is $\sum_{T: m_T = \min} u_T(0) > 0$. Therefore $\min \deg_q \chi^\lambda(\Omega) = \min_T m_T = \min_T \text{ord } p_T$.

Finally $\text{ord } p_T = \min\{\sum_k \text{cost}_k : \text{closed walks from } T\}$: the order of a sum of products is at least the minimum of the sums of orders, with equality because the minimal-order products have positive leading coefficients (again Hoefsmit positivity: each factor’s leading coefficient at its order is > 0 , see Lemma 8, so the minimal-order contributions are all positive and cannot cancel). Minimising over T gives the claim. $\square$

Remark 12. What Lemma 11 actually needs is only (i) every entry of $\rho^\lambda(T_i + 1)$ is in $\mathbb{Q}_{\geq 0}(q)$, which is [[hoefsmit-positivity]] (verified for $|\lambda| \leq 14$ and proved in the $b' = 1$ basis), and (ii) the leading low-order coefficient of each entry in Lemma 8 is positive, which is explicit there ($+1, +1, +1, +1$ as $q \rightarrow 0$). Both hold.

This proves Theorem 2. $\square$

4 Proof of Theorem 3 (upper bound via all-stay)

4.1 The all-stay walk

Fix a tableau U and consider the *all-stay* walk $U_0 = \dots = U_L = U$. It is a valid closed walk iff no step is forbidden, i.e. iff U has *no two consecutive integers in the same column* (otherwise some letter i with $i, i+1$ in a column appears in $\mathbf{w}_0$ and contributes the zero entry). When valid, its cost is, by Definition 9,

$$\text{cost}_{\text{stay}}(U) = \sum_{i=1}^{n-1} \mu(i) [c_U(i+1) < c_U(i)], \quad \mu(i) = \#\{k : i_k = i\} = n - i, \quad (4)$$

because letter i occurs in groups $G_{i+1}, \dots, G_n$, i.e. $n - i$ times.

4.2 The row–superstandard tableau

Let T_{rs} be the row–superstandard tableau (rows filled $1, 2, 3, \dots$ left to right, top to bottom). In T_{rs} the content strictly increases by 1 along each row, so the only content–descents $c(i+1) < c(i)$ occur at row breaks: when $i = P_r := \lambda_0 + \dots + \lambda_r$ is the last entry of row r (content $\lambda_r - 1 - r$) and $i+1 = P_r + 1$ begins row $r+1$ (content $-(r+1)$). The content drop there is $-(r+1) - (\lambda_r - 1 - r) = -\lambda_r \leq -1$, equal to -1 iff $\lambda_r = 1$.

Proposition 13. *If λ has at most one part equal to 1, then the all–stay walk at T_{rs} is valid and $\text{cost}_{\text{stay}}(T_{\text{rs}}) = n(\lambda)$. Hence $\text{mincost}(\lambda) \leq n(\lambda)$.*

Proof. Validity. A same–column consecutive pair would force a row break with content drop exactly -1 , i.e. $\lambda_r = 1$ for some $r \leq \ell-2$ (an internal row of length 1). Since the parts are weakly decreasing, having a length–1 part before the last row forces at least two parts equal to 1. By hypothesis there is at most one such part (necessarily the last row), so no internal row has length 1 and the walk is valid.

Cost. By (4) and the description of content–descents,

$$\text{cost}_{\text{stay}}(T_{\text{rs}}) = \sum_{r=0}^{\ell-2} (n - P_r) = \sum_{r=0}^{\ell-2} \sum_{s=r+1}^{\ell-1} \lambda_s = \sum_{s=1}^{\ell-1} \lambda_s \cdot \#\{r : 0 \leq r \leq s-1\} = \sum_{s=1}^{\ell-1} s \lambda_s = n(\lambda).$$

The middle step uses $n - P_r = \sum_{s>r} \lambda_s$. Combining with Lemma 11/Theorem 2 gives the bound. $\square$

Corollary 14 (two–row). *For $\lambda = (a, b)$ with $a \geq b \geq 1$ and $\chi^\lambda(\Omega) \neq 0$ (so $a \geq 2$), $\text{mincost}(\lambda) \leq b = n(\lambda)$.*

Proof. A two–row shape has at most one part equal to 1 (only when $b = 1$), so Proposition 13 applies; $n((a, b)) = b$. $\square$

5 The lower bound: ballot/leg reformulations and the charge statement

We record the explicit form of mincost that makes Conjecture 4 a concrete combinatorial statement, and prove the matching upper bound is sharp in the cases tested.

5.1 Two rows: the ballot model

Encode a two–row SYT of shape (a, b) by its row word $w \in \{U, D\}^n$: $w_v = U$ if v is in the top row, D if in the bottom row. Standardness $\Leftrightarrow w$ is a *ballot sequence* (every prefix has $\#U \geq \#D$), and there are a U’s, b D’s. The k –th U sits at content $k - 1$, the k –th D at content $k - 2$. A direct content computation gives the cost rule for letter i acting on positions $(i, i+1)$ of w :

pattern at $(i, i+1)$	status	cost
UU or DD	forced stay (same row)	0
DU	content ascent; optional swap to UD	0
UD with prefix excess $\#U - \#D \geq 2$	content descent; stay or swap	1
UD with prefix excess $\#U - \#D = 1$	same column	forbidden

Indeed if $w_i = U$ then the prefix excess through i is ≥ 1 , and one checks $c(i+1) - c(i) = i - 2k = -(\#U - \#D)$ where $k = \#U$ through i ; this is ≤ -2 , $= -1$, or (for $DU \geq 2$ exactly in the three rows above.

Thus $\text{mincost}((a, b))$ is the minimum number of cost-1 events in a closed walk that swaps only U - D adjacencies along the staircase word and never touches a height-1 peak. Corollary 14 exhibits the walk: all-stay at $w = U^a D^b$, whose only UD adjacency is at position a (excess $a \geq 2$), hit by the $b = \mu(a)$ occurrences of letter a , for total cost b . Conjecture 4 asserts no closed walk does better; this is verified for all two-row shapes with $n \leq 7$ (§6).

5.2 Hooks: the leg model

For a hook $\lambda = (a, 1^m)$ ($n = a + m$) the cells are the arm (row 0, contents $0, \dots, a-1$) and the leg (column 0, rows $1, \dots, m$, contents $-1, \dots, -m$). Here $n(\lambda) = \binom{m+1}{2} = \sum_{s=1}^m s$. When $m \geq 2$ the leg contains the consecutive same-column pairs, so all-stay at T_{rs} is *invalid*; the optimal walk (found by the dynamic program) instead performs $\binom{m}{2}$ -many “lift” swaps that temporarily raise a leg value into the arm to dodge a forbidden same-column step, then restores it, achieving cost exactly $\binom{m+1}{2}$. (Worked examples: $(3, 1, 1)$ and $(4, 1, 1)$ give cost $3 = \binom{3}{2}$; §6.)

5.3 The general charge statement

In both models the answer is $n(\lambda) = \sum_j \binom{\lambda'_j}{2}$, one unit per pair of cells in a common column. The natural *charge argument* is: assign to each column-pair of cells a distinct cost-1 step of the walk. The same-column constraint (forbidden steps) forces, for each column of length c , at least $\binom{c}{2}$ content-descent steps among the values that pass through that column. Making this injection precise — equivalently, exhibiting a layered shortest-path potential certifying $\text{mincost}(\lambda) \geq n(\lambda)$ — is the content of Conjecture 4. We have not completed it in general; see §7.

6 Computational verification

All computations use the validated Hoefsmit module `2026-04-25-biswal-chebyshev/compute_p_lambda.py` and the walk dynamic program `scratch/walklib.py`, `verify_all.py`.

- **Reciprocity (Thm 1).** For every non-vanishing shape with $n \leq 7$ (29 shapes), $q^{\binom{n}{2}} \chi^\lambda(\Omega)(q^{-1}) = \chi^\lambda(\Omega)(q)$ exactly: 29/29 PASS.
- **Reduction + bounds (Thms 2,3, Conj. 4).** For every non-vanishing shape with $n \leq 7$:

$$\text{mincost}(\lambda) = n(\lambda) = \sum_j \binom{\lambda'_j}{2} = \min \deg_q \chi^\lambda(\Omega), \quad \max \deg_q \chi^\lambda(\Omega) = \binom{n}{2} - n(\lambda),$$

with zero exceptions (29/29). E.g. $(3, 2, 1)$: $\min \deg = 4$, $\max \deg = 11$, span 15; $(2, 2, 2)$: 6 and 9, span 15; $(3, 3, 1, 1, 1)$: 12 and 24, span 36.

- The dynamic program reproduces the true min deg of the exact trace polynomial on all shapes in $\{(2, 2), (2, 2, 1), (3, 3, 1), (3, 3, 1, 1), (3, 3, 1, 1, 1), (3, 2, 1), (3, 2, 1, 1), (2, 1), (3, 1), (2, 2, 2), (3, 2), (4, 2), (3, 2, 2)\}$ and all of $n \leq 7$.

7 Gaps

1. **General lower bound (Conjecture 4).** $\text{mincost}(\lambda) \geq n(\lambda)$ is proved here only via the matching upper bound for shapes with ≤ 1 part equal to 1 (so for those shapes the *value*

$n(\lambda)$ is established as an upper bound, and equality is verified, but the inequality “no walk beats $n(\lambda)$ ” is not proved in closed form). The clean statement is the charge/injection of §5, or equivalently a shortest-path potential. Verified exactly for all 29 non-vanishing shapes with $n \leq 7$.

2. **General upper bound.** For λ with two or more parts equal to 1 (hooks $(a, 1^m)$, $m \geq 2$, and the like) the all-stay walk is invalid; the “lift-and-restore” construction of §5 achieves $n(\lambda)$ on every tested shape but is described, not proved, in general.
3. **Hoefsmit positivity** (Remark 12) is used as an input; it is proved in the $b' = 1$ basis and verified for $|\lambda| \leq 14$.

Theorem 1 (the principal result, and the part flagged as “the only real work”) is complete and unconditional. Theorem 2 is complete modulo the cited Hoefsmit positivity. Theorem 3 is complete and unconditional for the stated class. The remaining content of Conjecture 4 is the general lower bound.